

ZERO-TO-JOB-READY DATA ANALYST 2026

M01 - EXCEL FUNDAMENTALS

Beginner-first learner book | 9 lessons | visual-first practice

LEARN -> FOLLOW -> DO ALONE -> VALIDATE -> EXPLAIN -> MASTER

This is a clean regeneration, not a patched copy of the earlier learner book.

This book is your teacher

- You do not need to understand Excel words before they are taught here. Every lesson starts with why the skill matters, a picture in your head, and new words before we use them.
- Use the matching practice sheet while learning. When a visual source is assigned, watch only the exact segment, then immediately repeat the action in Excel. Watching without doing does not count.
- Independent work happens only after the skill has been demonstrated and guided once. A result is not accepted until you validate it and explain it in your own words.
- Protected reviews remain closed until a genuine attempt has been saved. After repair, close the review and solve a changed fresh task. Old answers are never your mastery evidence.
- There is no forced 25-minute study timer. Mentor study time is intended to use manual START STUDY / END STUDY real elapsed time.

Your normal lesson loop

1. Read/understand
2. Watch exact segment if assigned
3. Follow once
4. Do changed version alone
5. Validate
6. Explain back
7. Save evidence
8. Continue or repair

Contents / Index

EF1 Meet Excel: Workbook, Worksheet and Grid	6
EF2 Enter, Edit, Save and Structure Data	9
EF3 Data Types, Number Formats and Clean Layout	12
EF4 Sort and Filter Safely	15
EF5 Formula Basics and Arithmetic	18
EF6 Essential Functions: SUM, AVERAGE, MIN, MAX and COUNT	21
EF7 Cell References: Relative and Absolute	24
EF8 Excel Tables	27
EF9 Basic Charts for Clear Comparison	30
End-of-module mastery checklist	33
Revision / quick reference	34

Module prerequisite and tool map

0. Prerequisite

M00 Foundation mastered enough that you can state row grain, keep a raw/source concept separate from working analysis, validate before claiming success, and save evidence. No prior Excel skill is assumed.

Tools

- Laptop/desktop Excel. Excel 2016+ is sufficient for these fundamentals; Microsoft 365 is also fine.
- C1_M01_EXCEL_FUNDAMENTALS_CANONICAL_PRACTICE_WORKBOOK_RC1.xlsx.
- This learner book PDF. Open the matching lesson and practice sheet side-by-side.
- Protected Practice Review - closed until saved attempt.
- M01 Evidence_Log and Mastery_Tracker sheets inside the practice workbook.

What M01 intentionally does NOT teach yet

- No XLOOKUP/VLOOKUP yet.
- No PivotTables yet.
- No Power Query yet.
- No advanced chart design.
- No macros/VBA.

Exact study, practice and Gate route

- Study EF1. If a video is assigned, watch only its exact segment and then perform the follow-along in the matching practice sheet.
- Save a genuine attempt in Evidence_Log before opening any protected review.
- Close the review after repair. Complete the changed fresh retry without copying.
- Repeat through EF9. Then complete M-XLF Integration Mini-Lab mostly independently.
- Attempt Gate A. A clean competency pass ends the Gate.
- Only after failed A + Review A + targeted repair + explicit Mentor assignment may you attempt fresh Gate B.
- Only after failed B + Review B + targeted repair + explicit assignment may you attempt fresh Gate C.

Competency pass rule

- No average-percentage shortcut.
- Critical failure in formulas/references, record integrity, validation, Table creation or explanation blocks the pass.
- Mastery = changed task + validation + explain-back + saved evidence.

EF1 - Meet Excel: Workbook, Worksheet and Grid

Practice sheet: P-EF1_Navigation

1. Why this lesson matters

Excel is useful only when the workbook remains understandable, change-safe and auditable. The goal is not to memorize clicks; it is to produce a result that still makes sense when the data or question changes.

2. What you are learning

Meet Excel: Workbook, Worksheet and Grid at working independence: perform the core task, validate it, and explain what could go wrong.

3. What you already need / prerequisite

M00 Foundation only. No prior Excel skill is assumed.

4. Picture it in your head

Think of the workbook as the file, worksheets as tabs/pages inside it, and cells as addresses on a grid. The active cell is where your next entry will go.

5. New words before we use them

Workbook = Excel file. Worksheet = one tab/grid. Cell = one box. Cell address = column letter + row number, such as D7. Active cell = selected cell. Name Box = shows the active address. Formula Bar = shows or edits the selected cell content/formula.

6. Watch / visual source

RECOMMENDED visual demonstration. Excel Tutorial for Beginners: Learn it FAST | segment 1:15-4:09. Why watch: see the exact Excel action before you reproduce it. Internal fallback: this lesson fully explains and demonstrates the same skill, so a video outage never blocks mastery.

[Open verified/corroborated source](#)

EF1 - Teacher demonstration and recovery

7. Follow with me once - fully worked teacher demonstration

Open a workbook, switch between two worksheets, click C5, confirm C5 in the Name Box, type North, and inspect the value in the Formula Bar.

8. What you should see / be able to say

You can name workbook vs worksheet correctly, the requested tabs exist, E9 on Summary contains Ready after reopen, and you can point to Ribbon, Name Box and Formula Bar.

9. Why it worked

It works because the Excel action matches the business meaning and the validation checks the result rather than trusting appearance. The workbook keeps inputs, calculations and interpretation connected.

10. Common beginner mistakes

Typing before checking the active cell and placing data in the wrong address. | Confusing workbook (file) with worksheet (tab). | Editing the wrong worksheet because the tab was not checked first.

11. If you get stuck - recovery ladder

If typing lands in the wrong place, stop and verify the active-cell address in the Name Box before editing. If you are on the wrong tab, switch worksheets before changing data. Recovery ladder: 1) stop changing cells; 2) restate the intended result; 3) inspect the active sheet/range/formula; 4) undo to the last known-good state if needed; 5) reveal only the smallest hint; 6) use protected review only after a saved attempt.

EF1 - Do alone, validate and master

12. Now you do a small version alone

Create a fresh workbook with sheets Sales, Customers and Summary. Use the Name Box to jump to E9 on Summary, type Ready, save and reopen.

13. Validate your result

You can name workbook vs worksheet correctly, the requested tabs exist, E9 on Summary contains Ready after reopen, and you can point to Ribbon, Name Box and Formula Bar.

14. Teach it back in your own words

Without the lesson open, explain meet excel: workbook, worksheet and grid in 45-60 seconds. State what you did, why it was safe, one validation check and one common failure.

15. Protected review + changed fresh retry

Save a genuine attempt in Evidence_Log first. Then, if needed, open the protected review for EF1. Close it after repair and complete a changed retry. Do not overwrite the earlier attempt; append the retry as a new evidence row.

16. Evidence / mastery + exact next route

Save the workbook, one screenshot or cell/range reference showing the result, one validation/sanity check, one failure/recovery note, confidence 1-5, and a short own-words explanation. PASS with evidence -> continue. PARTIAL/UNCERTAIN -> targeted repair + fresh changed retry. FAIL -> repeat only the weak skill before continuing.

EF2 - Enter, Edit, Save and Structure Data

Practice sheet: P-EF2_Structure

1. Why this lesson matters

Excel is useful only when the workbook remains understandable, change-safe and auditable. The goal is not to memorize clicks; it is to produce a result that still makes sense when the data or question changes.

2. What you are learning

Enter, Edit, Save and Structure Data at working independence: perform the core task, validate it, and explain what could go wrong.

3. What you already need / prerequisite

M00 Foundation plus the previous Excel Fundamentals lesson.

4. Picture it in your head

A clean analyst dataset is rectangular: one header row, one field per column, one record per row at a stated grain, and no decorative blank rows/merged cells inside the data block.

5. New words before we use them

Field = one attribute/column. Record = one row/entity occurrence. Grain = what one row represents. Raw source = received original. Working copy = editable copy used for analysis.

6. Watch / visual source

RECOMMENDED visual demonstration. Excel Tutorial for Beginners: Learn it FAST | segment 4:09-8:20. Why watch: see the exact Excel action before you reproduce it. Internal fallback: this lesson fully explains and demonstrates the same skill, so a video outage never blocks mastery.

[Open verified/corroborated source](#)

EF2 - Teacher demonstration and recovery

7. Follow with me once - fully worked teacher demonstration

Build a six-column order block: Order ID, Order Date, Region, Product, Quantity, Unit Price. Enter five records and state: one row = one order line.

8. What you should see / be able to say

There is exactly one header row, one meaning per column, consistent row grain, no blank decorative rows inside the block, and the row concept is preserved separately from later calculations.

9. Why it worked

It works because the Excel action matches the business meaning and the validation checks the result rather than trusting appearance. The workbook keeps inputs, calculations and interpretation connected.

10. Common beginner mistakes

Using two header rows or merged cells inside the data block. | Mixing totals or notes into raw records. | Not being able to state what one row represents.

11. If you get stuck - recovery ladder

If your sheet has titles/totals mixed into raw records, move reporting text outside the data block. If you cannot state what one row means, stop before formulas and define the grain first. Recovery ladder: 1) stop changing cells; 2) restate the intended result; 3) inspect the active sheet/range/formula; 4) undo to the last known-good state if needed; 5) reveal only the smallest hint; 6) use protected review only after a saved attempt.

EF2 - Do alone, validate and master

12. Now you do a small version alone

Create a new five-record inventory-receipt table with Receipt ID, Date, Supplier, Item, Quantity and Unit Cost. No merged cells, multiple header rows or totals inside the raw block.

13. Validate your result

There is exactly one header row, one meaning per column, consistent row grain, no blank decorative rows inside the block, and the raw concept is preserved separately from later calculations.

14. Teach it back in your own words

Without the lesson open, explain enter, edit, save and structure data in 45-60 seconds. State what you did, why it was safe, one validation check and one common failure.

15. Protected review + changed fresh retry

Save a genuine attempt in Evidence_Log first. Then, if needed, open the protected review for EF2. Close it after repair and complete a changed retry. Do not overwrite the earlier attempt; append the retry as a new evidence row.

16. Evidence / mastery + exact next route

Save the workbook, one screenshot or cell/range reference showing the result, one validation/sanity check, one failure/recovery note, confidence 1-5, and a short own-words explanation. PASS with evidence -> continue. PARTIAL/UNCERTAIN -> targeted repair + fresh changed retry. FAIL -> repeat only the weak skill before continuing.

EF3 - Data Types, Number Formats and Clean Layout

Practice sheet: P-EF3_Formatting

1. Why this lesson matters

Excel is useful only when the workbook remains understandable, change-safe and auditable. The goal is not to memorize clicks; it is to produce a result that still makes sense when the data or question changes.

2. What you are learning

Data Types, Number Formats and Clean Layout at working independence: perform the core task, validate it, and explain what could go wrong.

3. What you already need / prerequisite

M00 Foundation plus the previous Excel Fundamentals lesson.

4. Picture it in your head

Excel stores a value and separately controls how it is displayed. 0.10 can display as 10%; a date is stored as a serial value; an ID may need Text so leading zeroes are preserved.

5. New words before we use them

Stored value = underlying value Excel calculates with. Number format = display rule. Text/label = identifier not meant for arithmetic. Percentage = displayed proportion, usually stored as decimal.

6. Watch / visual source

BOOK-LED. No external video is required for this lesson. The internal demonstration below is the canonical visual/step fallback.

EF3 - Teacher demonstration and recovery

7. Follow with me once - fully worked teacher demonstration

Format a valid Excel date as a readable date, 1250 as currency/number, 0.10 as 10%, and 00101 as an ID whose leading zeroes are preserved.

8. What you should see / be able to say

Changing a number format changes appearance without changing the underlying numeric value. IDs preserve required leading zeroes. Dates sort as dates rather than arbitrary text.

9. Why it worked

It works because the Excel action matches the business meaning and the validation checks the result rather than trusting appearance. The workbook keeps inputs, calculations and interpretation connected.

10. Common beginner mistakes

Typing '10%' as text rather than storing a usable numeric percentage. | Assuming formatting can repair a truly text value. | Treating an ID like a quantity and losing leading zeroes.

11. If you get stuck - recovery ladder

If 10% was typed as text or a number format does not fix the value, inspect the Formula Bar and stored value. Formatting cannot convert every bad text value into a real number/date. Recovery ladder: 1) stop changing cells; 2) restate the intended result; 3) inspect the active sheet/range/formula; 4) undo to the last known-good state if needed; 5) reveal only the smallest hint; 6) use protected review only after a saved attempt.

EF3 - Do alone, validate and master

12. Now you do a small version alone

Format a fresh subscription table containing Start Date, Monthly Fee, Renewal Rate, Active? and Member ID. Explain which fields are values versus labels.

13. Validate your result

Changing a number format changes appearance without changing the underlying numeric value. IDs preserve required leading zeroes. Dates sort as dates rather than arbitrary text.

14. Teach it back in your own words

Without the lesson open, explain data types, number formats and clean layout in 45-60 seconds. State what you did, why it was safe, one validation check and one common failure.

15. Protected review + changed fresh retry

Save a genuine attempt in Evidence_Log first. Then, if needed, open the protected review for EF3. Close it after repair and complete a changed retry. Do not overwrite the earlier attempt; append the retry as a new evidence row.

16. Evidence / mastery + exact next route

Save the workbook, one screenshot or cell/range reference showing the result, one validation/sanity check, one failure/recovery note, confidence 1-5, and a short own-words explanation. PASS with evidence -> continue. PARTIAL/UNCERTAIN -> targeted repair + fresh changed retry. FAIL -> repeat only the weak skill before continuing.

EF4 - Sort and Filter Safely

Practice sheet: P-EF4_SortFilter

1. Why this lesson matters

Excel is useful only when the workbook remains understandable, change-safe and auditable. The goal is not to memorize clicks; it is to produce a result that still makes sense when the data or question changes.

2. What you are learning

Sort and Filter Safely at working independence: perform the core task, validate it, and explain what could go wrong.

3. What you already need / prerequisite

M00 Foundation plus the previous Excel Fundamentals lesson.

4. Picture it in your head

Sorting changes row order; filtering temporarily hides rows. The record must move as a whole row. Sorting only one column can detach values from the correct record.

5. New words before we use them

Sort = reorder records. Filter = show only rows meeting conditions while keeping hidden rows. Record integrity = all fields for one record stay together.

6. Watch / visual source

BOOK-LED. No external video is required for this lesson. The internal demonstration below is the canonical visual/step fallback.

EF4 - Teacher demonstration and recovery

7. Follow with me once - fully worked teacher demonstration

Filter an order list to Cash only, sort Amount high-to-low, identify the top visible Order ID, then clear the filter and confirm all rows return.

8. What you should see / be able to say

The top filtered record is correct, full-row relationships remain intact, and clearing the filter restores the original number of records.

9. Why it worked

It works because the Excel action matches the business meaning and the validation checks the result rather than trusting appearance. The workbook keeps inputs, calculations and interpretation connected.

10. Common beginner mistakes

Sorting only one column and breaking row relationships. | Forgetting to clear the filter and believing hidden rows were deleted. | Reading the visible top row without verifying the filter condition.

11. If you get stuck - recovery ladder

If values no longer match their IDs/customers, undo immediately and restore from the working/raw copy. Always select the whole data region or use a Table before sorting. Recovery ladder: 1) stop changing cells; 2) restate the intended result; 3) inspect the active sheet/range/formula; 4) undo to the last known-good state if needed; 5) reveal only the smallest hint; 6) use protected review only after a saved attempt.

EF4 - Do alone, validate and master

12. Now you do a small version alone

On a fresh service-ticket list, filter Priority = High, sort Open Days descending, identify the oldest High ticket, then clear the filter and confirm every record returns.

13. Validate your result

The top filtered record is correct, full-row relationships remain intact, and clearing the filter restores the original number of records.

14. Teach it back in your own words

Without the lesson open, explain sort and filter safely in 45-60 seconds. State what you did, why it was safe, one validation check and one common failure.

15. Protected review + changed fresh retry

Save a genuine attempt in Evidence_Log first. Then, if needed, open the protected review for EF4. Close it after repair and complete a changed retry. Do not overwrite the earlier attempt; append the retry as a new evidence row.

16. Evidence / mastery + exact next route

Save the workbook, one screenshot or cell/range reference showing the result, one validation/sanity check, one failure/recovery note, confidence 1-5, and a short own-words explanation. PASS with evidence -> continue. PARTIAL/UNCERTAIN -> targeted repair + fresh changed retry. FAIL -> repeat only the weak skill before continuing.

EF5 - Formula Basics and Arithmetic

Practice sheet: P-EF5_Formulas

1. Why this lesson matters

Excel is useful only when the workbook remains understandable, change-safe and auditable. The goal is not to memorize clicks; it is to produce a result that still makes sense when the data or question changes.

2. What you are learning

Formula Basics and Arithmetic at working independence: perform the core task, validate it, and explain what could go wrong.

3. What you already need / prerequisite

M00 Foundation plus the previous Excel Fundamentals lesson.

4. Picture it in your head

A formula is a live instruction beginning with =. Refer to input cells instead of typing hard-coded answers so results recalculate when inputs change.

5. New words before we use them

Formula = expression beginning with =. Cell reference = address used as an input. Operator = +, -, *, /. Hard-coded number = typed constant inside logic that should have come from a cell.

6. Watch / visual source

RECOMMENDED visual demonstration. Excel Tutorial for Beginners: Learn it FAST | segment 15:46-21:00. Why watch: see the exact Excel action before you reproduce it. Internal fallback: this lesson fully explains and demonstrates the same skill, so a video outage never blocks mastery.

[Open verified/corroborated source](#)

EF5 - Teacher demonstration and recovery

7. Follow with me once - fully worked teacher demonstration

For Quantity in B12 and Unit Price in C12, Revenue = B12*C12. If Cost is D12, Profit = E12-D12. Margin = F12/E12 and display as Percentage.

8. What you should see / be able to say

Formulas reference the intended cells, copied rows adjust correctly, outputs change after an input changes, and one row can be recomputed manually.

9. Why it worked

It works because the Excel action matches the business meaning and the validation checks the result rather than trusting appearance. The workbook keeps inputs, calculations and interpretation connected.

10. Common beginner mistakes

Typing a calculated answer instead of a formula. | Hard-coding input numbers inside formulas when the inputs already exist in cells. | Copying a formula without manually checking one row.

11. If you get stuck - recovery ladder

If copied formulas point to the wrong row, inspect the Formula Bar. If a displayed percentage is huge, check whether you divided in the correct direction and formatted the result as Percentage. Recovery ladder: 1) stop changing cells; 2) restate the intended result; 3) inspect the active sheet/range/formula; 4) undo to the last known-good state if needed; 5) reveal only the smallest hint; 6) use protected review only after a saved attempt.

EF5 - Do alone, validate and master

12. Now you do a small version alone

On a fresh invoice block, calculate Line Total = Units * Rate and Balance = Line Total - Paid Amount. Copy formulas and test a changed input.

13. Validate your result

Formulas reference the intended cells, copied rows adjust correctly, outputs change after an input changes, and one row can be recomputed manually.

14. Teach it back in your own words

Without the lesson open, explain formula basics and arithmetic in 45-60 seconds. State what you did, why it was safe, one validation check and one common failure.

15. Protected review + changed fresh retry

Save a genuine attempt in Evidence_Log first. Then, if needed, open the protected review for EF5. Close it after repair and complete a changed retry. Do not overwrite the earlier attempt; append the retry as a new evidence row.

16. Evidence / mastery + exact next route

Save the workbook, one screenshot or cell/range reference showing the result, one validation/sanity check, one failure/recovery note, confidence 1-5, and a short own-words explanation. PASS with evidence -> continue. PARTIAL/UNCERTAIN -> targeted repair + fresh changed retry. FAIL -> repeat only the weak skill before continuing.

EF6 - Essential Functions: SUM, AVERAGE, MIN, MAX and COUNT

Practice sheet: P-EF6_Functions

1. Why this lesson matters

Excel is useful only when the workbook remains understandable, change-safe and auditable. The goal is not to memorize clicks; it is to produce a result that still makes sense when the data or question changes.

2. What you are learning

Essential Functions: SUM, AVERAGE, MIN, MAX and COUNT at working independence: perform the core task, validate it, and explain what could go wrong.

3. What you already need / prerequisite

M00 Foundation plus the previous Excel Fundamentals lesson.

4. Picture it in your head

Functions summarize a range. SUM totals values; AVERAGE gives the arithmetic mean; MIN/MAX find extremes; COUNT counts numeric cells - not unique customers or valid business records.

5. New words before we use them

Function = built-in calculation. Range = cells such as B12:B19. COUNT = numeric-cell count. AVERAGE = sum divided by numeric count.

6. Watch / visual source

RECOMMENDED visual demonstration. Excel Tutorial for Beginners: Learn it FAST | segment 21:29-22:38. Why watch: see the exact Excel action before you reproduce it. Internal fallback: this lesson fully explains and demonstrates the same skill, so a video outage never blocks mastery.

[Open verified/corroborated source](#)

EF6 - Teacher demonstration and recovery

7. Follow with me once - fully worked teacher demonstration

For B12:B19, enter SUM, AVERAGE, MIN, MAX and COUNT. Read each function in plain English before trusting the result.

8. What you should see / be able to say

The range boundaries are correct; COUNT matches the number of numeric cells; the total reconciles to an independent subtotal/manual check.

9. Why it worked

It works because the Excel action matches the business meaning and the validation checks the result rather than trusting appearance. The workbook keeps inputs, calculations and interpretation connected.

10. Common beginner mistakes

Using the wrong range endpoint. | Assuming COUNT proves uniqueness or data quality. | Reading AVERAGE without checking whether the intended rows are included.

11. If you get stuck - recovery ladder

If a function ignores expected cells, inspect the range endpoints and whether values are numeric. Do not widen the range blindly if headers or unrelated totals are nearby. Recovery ladder: 1) stop changing cells; 2) restate the intended result; 3) inspect the active sheet/range/formula; 4) undo to the last known-good state if needed; 5) reveal only the smallest hint; 6) use protected review only after a saved attempt.

EF6 - Do alone, validate and master

12. Now you do a small version alone

On a fresh weekly-orders range, calculate all five functions from memory and manually verify the SUM using two smaller subtotals.

13. Validate your result

The range boundaries are correct; COUNT matches the number of numeric cells; the total reconciles to an independent subtotal/manual check.

14. Teach it back in your own words

Without the lesson open, explain essential functions: sum, average, min, max and count in 45-60 seconds. State what you did, why it was safe, one validation check and one common failure.

15. Protected review + changed fresh retry

Save a genuine attempt in Evidence_Log first. Then, if needed, open the protected review for EF6. Close it after repair and complete a changed retry. Do not overwrite the earlier attempt; append the retry as a new evidence row.

16. Evidence / mastery + exact next route

Save the workbook, one screenshot or cell/range reference showing the result, one validation/sanity check, one failure/recovery note, confidence 1-5, and a short own-words explanation. PASS with evidence -> continue. PARTIAL/UNCERTAIN -> targeted repair + fresh changed retry. FAIL -> repeat only the weak skill before continuing.

EF7 - Cell References: Relative and Absolute

Practice sheet: P-EF7_References

1. Why this lesson matters

Excel is useful only when the workbook remains understandable, change-safe and auditable. The goal is not to memorize clicks; it is to produce a result that still makes sense when the data or question changes.

2. What you are learning

Cell References: Relative and Absolute at working independence: perform the core task, validate it, and explain what could go wrong.

3. What you already need / prerequisite

M00 Foundation plus the previous Excel Fundamentals lesson.

4. Picture it in your head

A relative reference moves when copied because it describes a relative position. An absolute reference such as \$H\$11 stays fixed because that business setting is shared by every row.

5. New words before we use them

Relative reference = A1. Absolute reference = \$A\$1. Mixed reference = \$A1 or A\$1. Copy/fill = repeat a formula while Excel adjusts relative parts.

6. Watch / visual source

RECOMMENDED visual demonstration. Learn ALL the useful Excel concepts in 40 mins - Tables, Formulas, Pivots & Charts (current verified primary). Legacy optional companion retained in the source registry: How to use Excel formula references - A1 vs. \$A\$1 vs \$A1 vs A\$1 explained. | segment 12:35-20:52. Why watch: see the exact Excel action before you reproduce it. Internal fallback: this lesson fully explains and demonstrates the same skill, so a video outage never blocks mastery.

[Open verified/corroborated source](#)

EF7 - Teacher demonstration and recovery

7. Follow with me once - fully worked teacher demonstration

If C13 is Price and H11 is one Discount Rate, use `=C13*$H$11`. Copy one row down: C13 should move to C14 while \$H\$11 remains fixed.

8. What you should see / be able to say

Copied formulas use each row input but keep the one shared setting fixed. You can predict the next formula before copying.

9. Why it worked

It works because the Excel action matches the business meaning and the validation checks the result rather than trusting appearance. The workbook keeps inputs, calculations and interpretation connected.

10. Common beginner mistakes

Forgetting the \$ signs on the shared rate. | Anchoring every reference so row inputs no longer move. | Copying first and only later trying to understand why results changed.

11. If you get stuck - recovery ladder

If the fixed setting changes row-by-row, add the correct \$ anchors. If everything is anchored, remove \$ from references that are supposed to move. Recovery ladder: 1) stop changing cells; 2) restate the intended result; 3) inspect the active sheet/range/formula; 4) undo to the last known-good state if needed; 5) reveal only the smallest hint; 6) use protected review only after a saved attempt.

EF7 - Do alone, validate and master

12. Now you do a small version alone

Use one Tax Rate setting in J5 for six invoice rows. Calculate Tax = Amount*\$J\$5 and Final = Amount+Tax. Explain why the rate is absolute.

13. Validate your result

Copied formulas use each row input but keep the one shared setting fixed. You can predict the next formula before copying.

14. Teach it back in your own words

Without the lesson open, explain cell references: relative and absolute in 45-60 seconds. State what you did, why it was safe, one validation check and one common failure.

15. Protected review + changed fresh retry

Save a genuine attempt in Evidence_Log first. Then, if needed, open the protected review for EF7. Close it after repair and complete a changed retry. Do not overwrite the earlier attempt; append the retry as a new evidence row.

16. Evidence / mastery + exact next route

Save the workbook, one screenshot or cell/range reference showing the result, one validation/sanity check, one failure/recovery note, confidence 1-5, and a short own-words explanation. PASS with evidence -> continue. PARTIAL/UNCERTAIN -> targeted repair + fresh changed retry. FAIL -> repeat only the weak skill before continuing.

EF8 - Excel Tables

Practice sheet: P-EF8_Tables

1. Why this lesson matters

Excel is useful only when the workbook remains understandable, change-safe and auditable. The goal is not to memorize clicks; it is to produce a result that still makes sense when the data or question changes.

2. What you are learning

Excel Tables at working independence: perform the core task, validate it, and explain what could go wrong.

3. What you already need / prerequisite

M00 Foundation plus the previous Excel Fundamentals lesson.

4. Picture it in your head

An Excel Table is a structured object, not merely colored cells. It has a defined data region, header/filter behavior, a name, and normally expands as new rows are added.

5. New words before we use them

Excel Table = structured range object. Table name = identifier such as PracticeOrders. Header row = field names. Calculated column = formula behavior applied through a Table column.

6. Watch / visual source

RECOMMENDED visual demonstration. Learn ALL the useful Excel concepts in 40 mins | segment 2:22-8:44. Why watch: see the exact Excel action before you reproduce it. Internal fallback: this lesson fully explains and demonstrates the same skill, so a video outage never blocks mastery.

[Open verified/corroborated source](#)

EF8 - Teacher demonstration and recovery

7. Follow with me once - fully worked teacher demonstration

Convert a clean range to a Table, confirm headers, name it DemoOrders, filter it, then add one row directly below and observe expansion.

8. What you should see / be able to say

The object is a real Excel Table with the required name, filters work, a new row belongs to the Table, and you can explain that a Table does not guarantee data quality.

9. Why it worked

It works because the Excel action matches the business meaning and the validation checks the result rather than trusting appearance. The workbook keeps inputs, calculations and interpretation connected.

10. Common beginner mistakes

Coloring a range and calling it an Excel Table. | Giving the Table an unclear/default name and losing track of it. | Believing Table structure automatically guarantees clean or correct data.

11. If you get stuck - recovery ladder

If you only changed colors/borders, you did not create a Table. Use Insert > Table or Ctrl+T on the whole clean range, confirm headers, then set the Table Name. Recovery ladder: 1) stop changing cells; 2) restate the intended result; 3) inspect the active sheet/range/formula; 4) undo to the last known-good state if needed; 5) reveal only the smallest hint; 6) use protected review only after a saved attempt.

EF8 - Do alone, validate and master

12. Now you do a small version alone

Convert a fresh customer-contact range to ContactsTable. Add one new record, filter one city, then clear the filter and explain three Table benefits.

13. Validate your result

The object is a real Excel Table with the required name, filters work, a new row belongs to the Table, and you can explain that a Table does not guarantee data quality.

14. Teach it back in your own words

Without the lesson open, explain excel tables in 45-60 seconds. State what you did, why it was safe, one validation check and one common failure.

15. Protected review + changed fresh retry

Save a genuine attempt in Evidence_Log first. Then, if needed, open the protected review for EF8. Close it after repair and complete a changed retry. Do not overwrite the earlier attempt; append the retry as a new evidence row.

16. Evidence / mastery + exact next route

Save the workbook, one screenshot or cell/range reference showing the result, one validation/sanity check, one failure/recovery note, confidence 1-5, and a short own-words explanation. PASS with evidence -> continue. PARTIAL/UNCERTAIN -> targeted repair + fresh changed retry. FAIL -> repeat only the weak skill before continuing.

EF9 - Basic Charts for Clear Comparison

Practice sheet: P-EF9_Charts

1. Why this lesson matters

Excel is useful only when the workbook remains understandable, change-safe and auditable. The goal is not to memorize clicks; it is to produce a result that still makes sense when the data or question changes.

2. What you are learning

Basic Charts for Clear Comparison at working independence: perform the core task, validate it, and explain what could go wrong.

3. What you already need / prerequisite

M00 Foundation plus the previous Excel Fundamentals lesson.

4. Picture it in your head

A chart is a visual sentence. Use bar/column for independent categories and line for ordered time. The chart should answer the question, not decorate the sheet.

5. New words before we use them

Category comparison = separate groups such as regions. Time trend = ordered dates/months. Chart title = statement of what is shown. 3D effects = usually unnecessary distortion for beginner reporting.

6. Watch / visual source

RECOMMENDED visual demonstration. Excel Tutorial for Beginners: Learn it FAST | segment 22:39-27:26. Why watch: see the exact Excel action before you reproduce it. Internal fallback: this lesson fully explains and demonstrates the same skill, so a video outage never blocks mastery.

[Open verified/corroborated source](#)

EF9 - Teacher demonstration and recovery

7. Follow with me once - fully worked teacher demonstration

Create a column chart for revenue by region. Then create a line chart for monthly revenue Jan-Jun and give it a meaningful title.

8. What you should see / be able to say

The chart source range is correct, category/time order is correct, title is meaningful, no unnecessary 3D effects are used, and your sentence describes rather than invents a cause.

9. Why it worked

It works because the Excel action matches the business meaning and the validation checks the result rather than trusting appearance. The workbook keeps inputs, calculations and interpretation connected.

10. Common beginner mistakes

Using a line chart for unordered categories. | Selecting the wrong range or swapping labels and values. | Writing a causal story when the chart only shows an observed pattern.

11. If you get stuck - recovery ladder

If categories/values are swapped or the chart tells the wrong relationship, check the source range and chart type. If the sentence claims a cause, rewrite it as an observed pattern only. Recovery ladder: 1) stop changing cells; 2) restate the intended result; 3) inspect the active sheet/range/formula; 4) undo to the last known-good state if needed; 5) reveal only the smallest hint; 6) use protected review only after a saved attempt.

EF9 - Do alone, validate and master

12. Now you do a small version alone

Create a column chart comparing five products by units sold. Then explain why a line chart would be misleading for unordered product categories.

13. Validate your result

The chart source range is correct, category/time order is correct, title is meaningful, no unnecessary 3D effects are used, and your sentence describes rather than invents a cause.

14. Teach it back in your own words

Without the lesson open, explain basic charts for clear comparison in 45-60 seconds. State what you did, why it was safe, one validation check and one common failure.

15. Protected review + changed fresh retry

Save a genuine attempt in Evidence_Log first. Then, if needed, open the protected review for EF9. Close it after repair and complete a changed retry. Do not overwrite the earlier attempt; append the retry as a new evidence row.

16. Evidence / mastery + exact next route

Save the workbook, one screenshot or cell/range reference showing the result, one validation/sanity check, one failure/recovery note, confidence 1-5, and a short own-words explanation. PASS with evidence -> continue. PARTIAL/UNCERTAIN -> targeted repair + fresh changed retry. FAIL -> repeat only the weak skill before continuing.

End-of-module mastery checklist

- EF1: Navigate workbook/worksheets/cells and identify Name Box / Formula Bar.
- EF2: Build a rectangular dataset and state one-row grain.
- EF3: Distinguish stored values from display formats; preserve IDs correctly.
- EF4: Sort/filter without breaking record integrity; restore all rows.
- EF5: Write arithmetic formulas from cell references and validate one row.
- EF6: Use SUM, AVERAGE, MIN, MAX and COUNT with correct ranges.
- EF7: Predict and use relative versus absolute references.
- EF8: Create and name a real Excel Table; explain what it does and does not guarantee.
- EF9: Choose a sensible basic chart and describe evidence without inventing causation.
- M-XLF: Combine the skills on changed data without lesson/review copying.

Gate readiness

Attempt Gate A only when the Mini-Lab and all weak EF competencies have fresh evidence.

A clean Gate A pass ends the Gate. B/C are remediation retests, not normal progression.

Revision / quick reference

Workbook / worksheet / cell	File / tab / grid address.
Grain	What one row represents.
Stored value vs format	The value drives calculation; the format controls display.
Safe sort/filter	Move whole records; clear filter and verify row count returns.
Formula	Starts with = and should reference cells when inputs already exist.
Five basics	SUM, AVERAGE, MIN, MAX, COUNT.
Relative / absolute	A1 moves when copied; \$A\$1 stays fixed.
Excel Table	Structured data object with headers, filters, name and expansion behavior.
Chart rule	Column/bar for categories; line for ordered time; describe before explaining.

Next route: Integration Mini-Lab -> Gate A -> M02 only after M01 competency pass.